

Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

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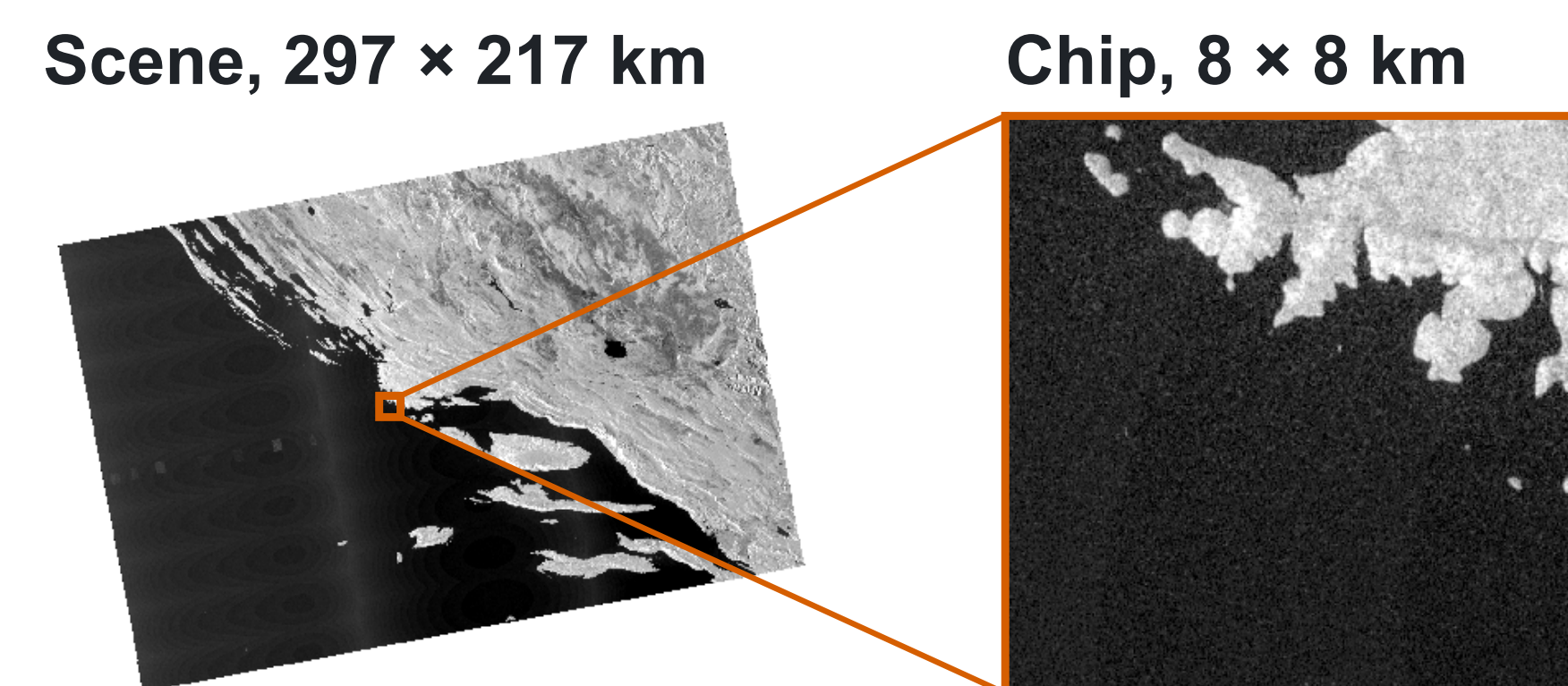
Johns Hopkins University

Answer: no. SAR pretraining beats optical in 6 of 16 comparisons and ImageNet in 3 of 16 (ImageNet wins 7); its clearest edge is the CNN with 12–28 labeled scenes.

Motivation

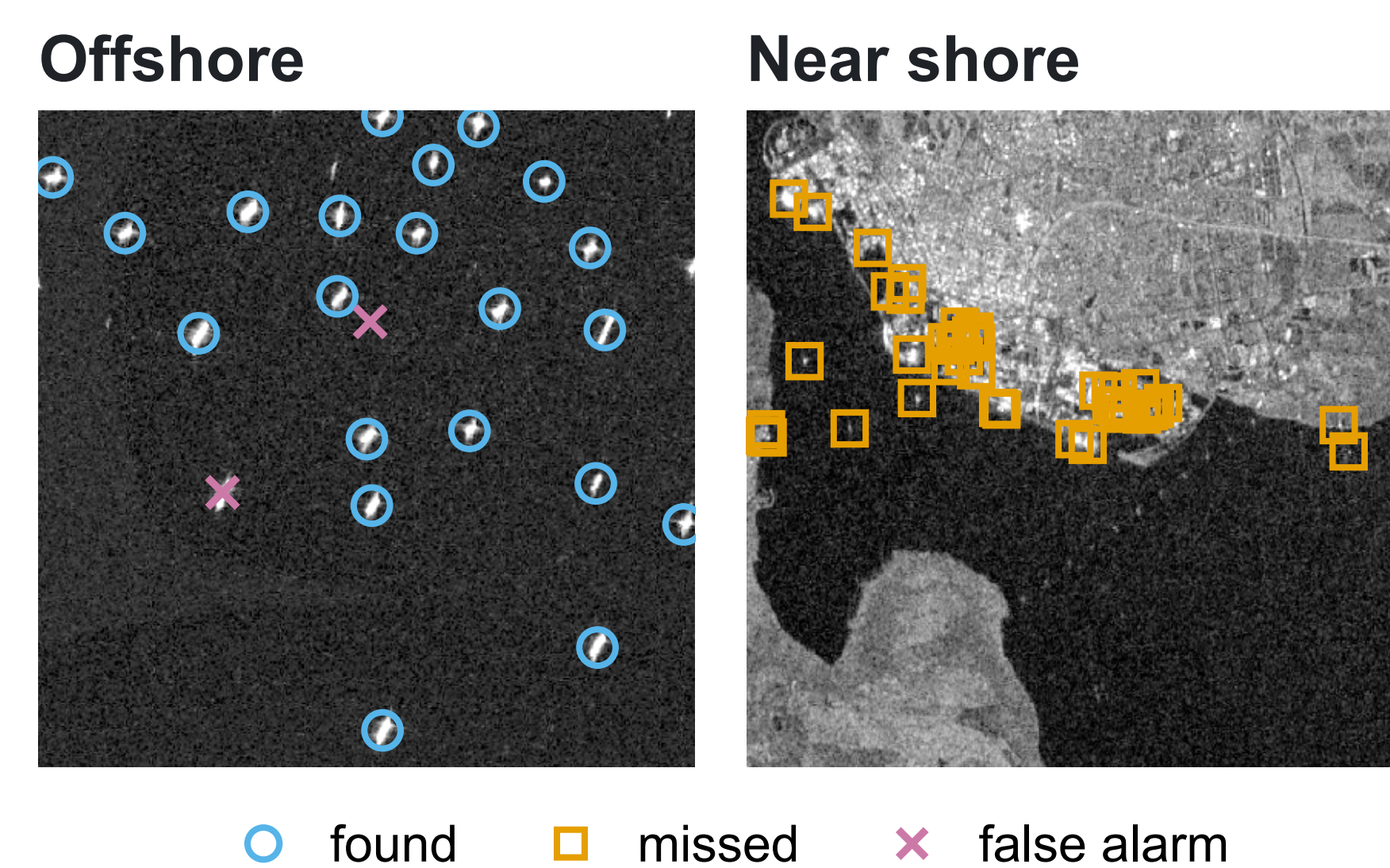
- **Dark vessels** have no matching AIS report. Sentinel-1 SAR [2] sees them at night and through cloud.
- Vessels fill a few pixels in a 25,000-pixel scene. Sea clutter and coastlines give strong false returns.
- **Human-verified labels are scarce.** Pretraining can lower the label cost, but which source domain helps most, and does the answer depend on the backbone?

Data: xView3-SAR [1]



- 150 frozen study scenes: **111 train, 23 dev, 16 test**. Nested budgets: $12 \subset 28 \subset 56 \subset 111$ training scenes.
- **Train, dev and test labels are machine-made:** AIS matching plus automated detection [1]. They miss vessels and are not complete ground truth.
- **50 near-shore, human-verified scenes**, opened once after test scoring: 8,642 vessels, 3,644 dark, 2,686 within 2 km of shore.

What the best detector finds



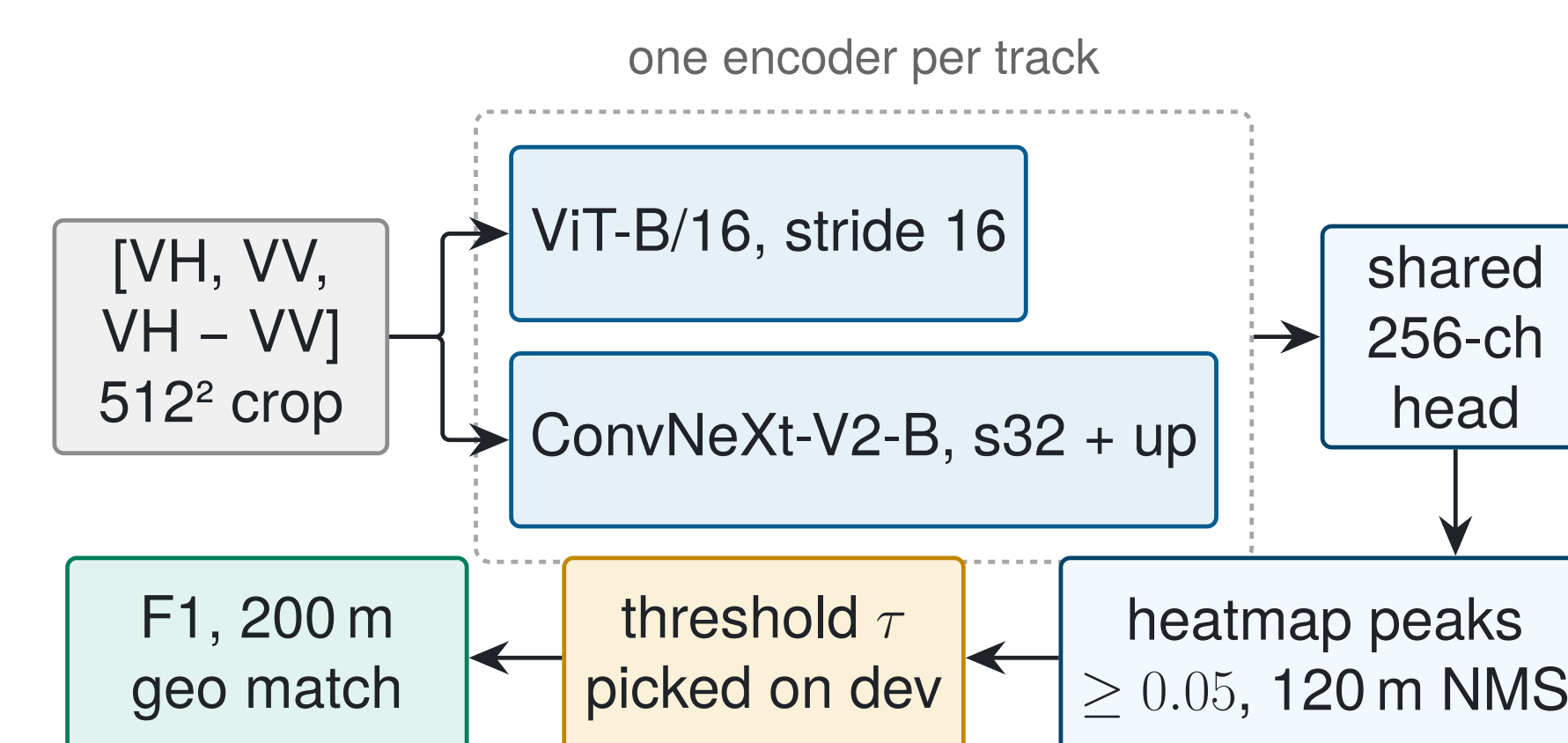
Best cell (ImageNet CNN, 111 scenes) on two 8×8 km verified crops (VH, dB): offshore it finds almost every vessel; in this harbor, none.

8 arms $\times$ 4 budgets = 32 cells

Role	ViT-B/16 [3]	ConvNeXt V2 [4]
Random	fresh init	fresh init
Optical	SatDINO [6] (fMoW-RGB)	BigEarthNet-S2 [8]
SAR	SARMAE [7] (SAR-1M)	BigEarthNet-S1 [8]
ImageNet	AugReg [5]	FCMAE [4]

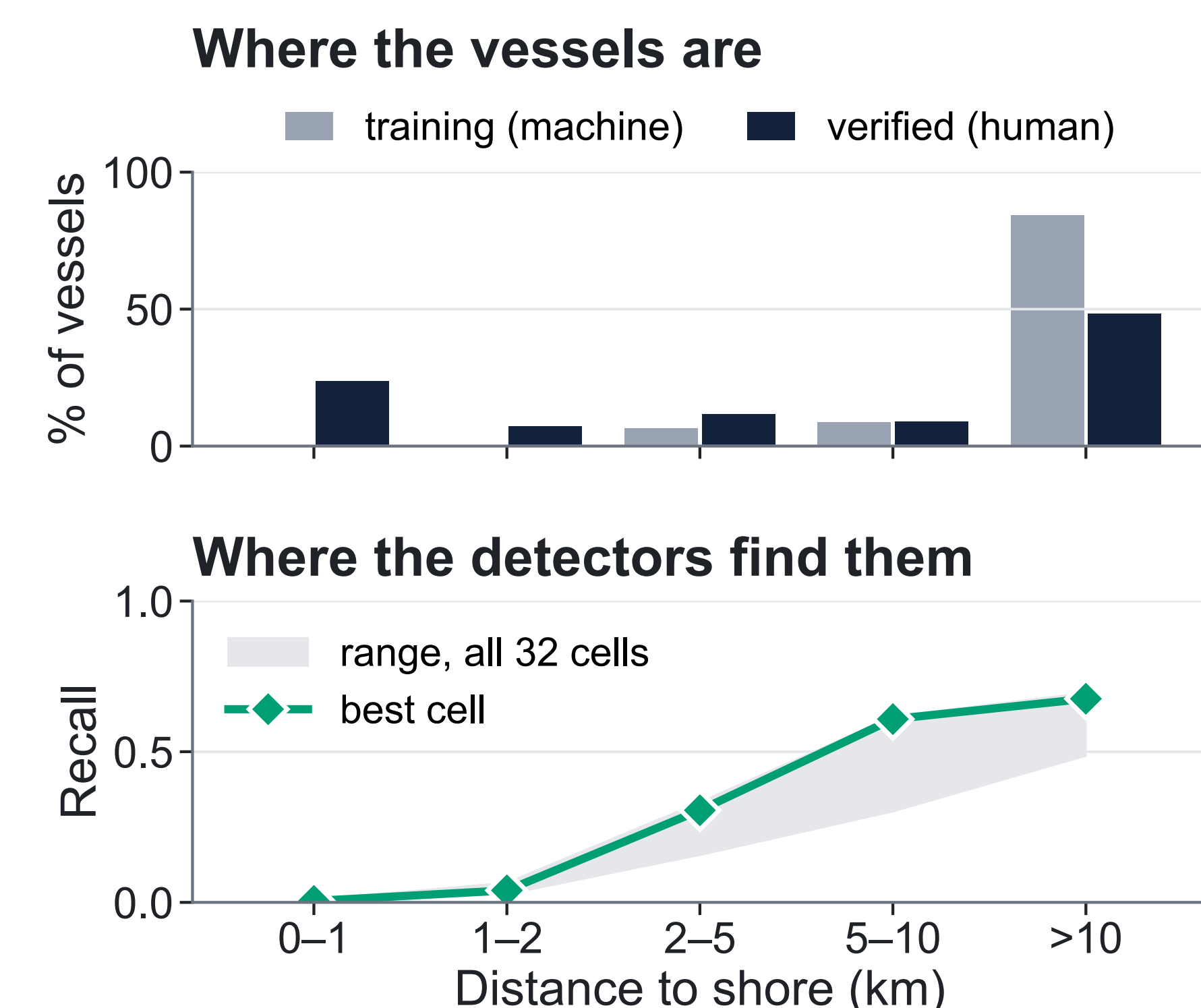
Only the initialization changes inside a track; we compare domains **within** an architecture.

One shared point detector



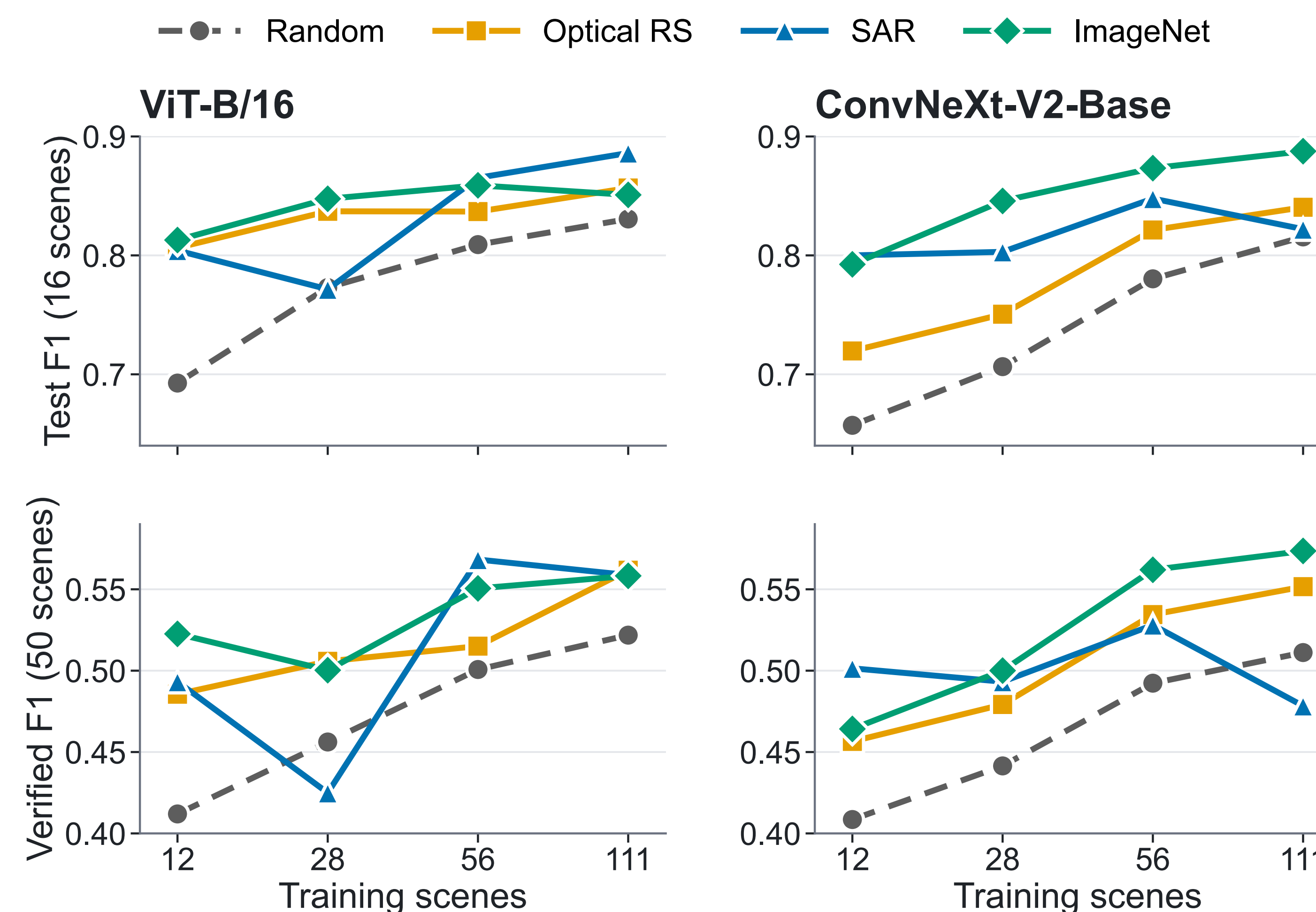
- CenterNet-style heatmap [9] with focal loss [10]; strict FP32, 567.6 H100 GPU-hours.
- Same optimizer, schedule, data order and seed (0) in every cell.
- 8 dev scenes pick each checkpoint and threshold; test and verified scenes are scored **once**.

Recall falls toward the coast



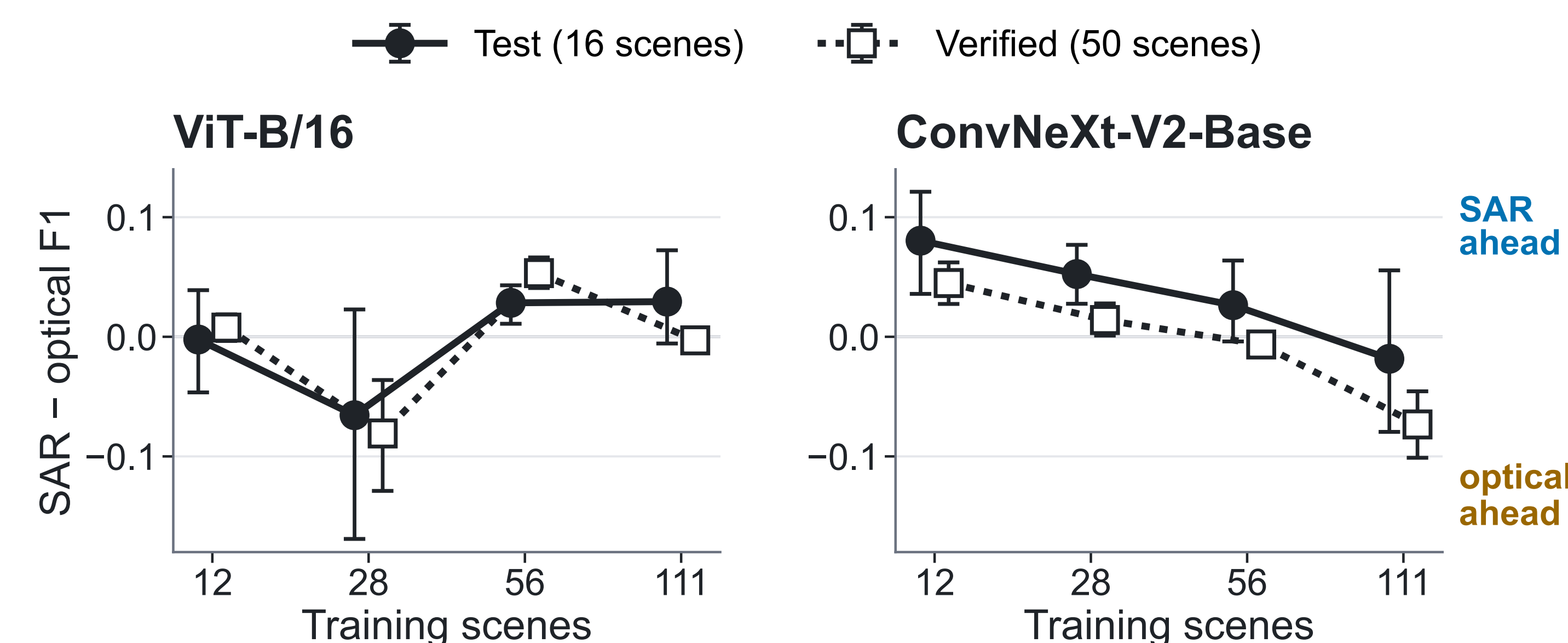
50 verified scenes; the band spans all 32 cells. Best-cell recall: 0.676 beyond 10 km, 0.005 within 1 km.

Pretraining helps most when labels are scarce



F1 with 200 m matching. Each of the 32 cells is scored once with the threshold it picked on 8 dev scenes. Top: 16 held-out test scenes. Bottom: 50 near-shore, human-verified scenes. Seed 0.

SAR vs. optical depends on backbone and budget



How to read: each point is SAR F1 minus optical F1 at one label budget. Above 0, SAR wins; below 0, optical wins. Solid: test scenes; dotted: verified scenes. A bar that does not cross 0 is a win; the 16 points here are the comparisons counted in the answer above. Bars are 95% intervals from 10,000 paired scene resamples and cover scene sampling only: retraining 28 unchanged cells moved test F1 by up to 0.043.

Near shore, every detector fails

0.026 vs 0.724 $\leq$ 0.021 0.4% vs 31%

best cell's verified F1: within 2 km vs offshore
near-shore recall, every cell
vessels within 2 km of shore: training vs verified labels
The machine-made training labels place almost no vessels near shore, and the detectors almost never fire there: the best cell puts 34 of its 4,302 predictions within 2 km. Precision stays at 0.639–0.909 in every cell.

Findings

12 / 12

pretrained-vs-random comparisons at 12 training scenes have a 95% interval above zero.

- 1 Pretraining pays off with few labels.** Gains at 12 scenes: +0.063 to +0.143 test F1, +0.048 to +0.111 verified F1.
- 2 SAR's edge over optical depends on backbone and budget.** CNN: +0.080 [+0.036, +0.121] test F1 at 12 scenes, but -0.073 [-0.101, -0.046] verified F1 at 111. ViT: the order flips between 28 and 56 scenes.
- 3 The input adapter matters.** With a seeded 3-channel stem instead of band slicing, BigEarthNet-S2 beats random at every budget (+0.025 to +0.063 test F1).
- 4 No detector works near shore:** recall ≤ 0.021 within 2 km in every cell. Training labels put 0.4% of vessels there; the verified set, 31%.

Limits

- One seed per cell; 8 dev scenes pick checkpoints and thresholds.
- Dev and test labels are machine-made, so test F1 measures agreement with them, not with every vessel.
- Two SAR test curves drop by more than 0.02 as labels grow. ViT-SAR at 28 scenes overfit after warm-up; early stopping kept epoch 4.
- Checkpoints differ in objective and corpus, not only domain.

Open questions & next steps

- How do geospatial foundation models pre-trained jointly on SAR and optical imagery fare against these single-source checkpoints?
- Does SAR pretraining help where optical cannot: near shore and for small dark vessels?
- **Next:** near-shore training labels, a shoreline input channel, several seeds, and matched pretraining objectives across sensors.